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Description - This material describes how to change the connectors on plastic fiber optic cables.

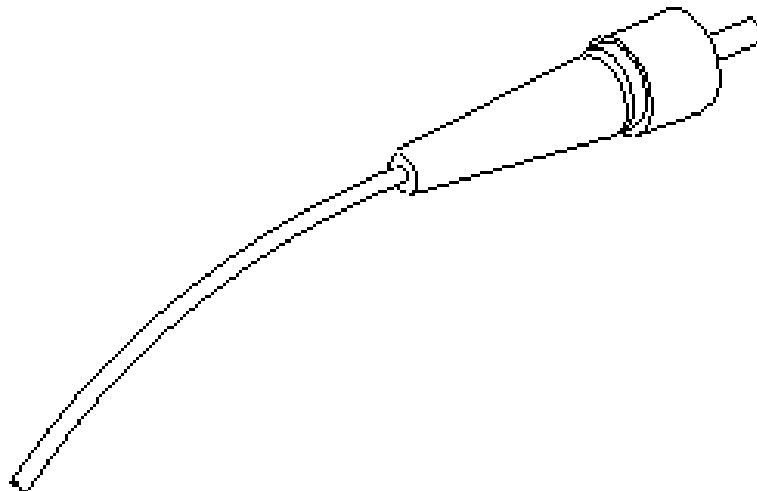
1- OVERVIEW

The connectors on **plastic** fiber optic cables can be replaced if damaged. The Fiber Optic Repair Kit, 46-301450G1 contains all the necessary materials to replace a damaged connector.

Note

The connectors on glass fiber optic cables (see Illustration L2950A) cannot be replaced using the kit listed in Table 1. At this time, there is no kit available to repair glass fiber optic cable connectors

Also see Illustration L2950A.



L2950A

GLASS FIBER OPTIC CONNECTOR
ILLUSTRATION L2950A

The end of the fiber optic cable to be repaired is removed from the magnet room, if possible. The bad connector is cut off with a side cutter, and the fiber optic cable is carefully stripped of its vinyl insulation. A crimp tool is used to attach a crimp ring and connector to the stripped fiber. Approximately one-half of the protruding fiber is cut off and the remaining fiber stub is made flush to the end of the connector with abrasive paper. The sanded fiber end is then polished, and the cable is put back into the system.

2- REQUIRED MATERIALS

The parts contained in the Fiber Optic Repair Kit, 46-301450G1, are listed in Table 1.

TABLE 1
FIBER OPTIC REPAIR KIT

Item	Description	Part Number	Quantity
1	Crimp Tool, Amp 90364-2	46-301451P1	1
2	Polishing Kit, HFBR-4595	46-301452P1	4
3	Connector (blue), Simplex, HFBR-4501 with crimp ring	2104238	1 (bag of 20)
4	Connector (gray), Simplex, HFBR-4511 with crimp ring	2104238-2	1 (bag of 20)
5	Connector (gray), Simplex, Latching, HFBR-4503 with crimp ring	2104238-3	1 (bag of 20)
6	Connector (blue), Simplex, Latching, HFBR-4513 with crimp ring	2104238-4	1 (bag of 20)
7	Connector (parchment), Duplex, HFBR-4506 with crimp ring	2104238-5	1 (bag of 20)
8	Connector (gray), Duplex, Latching, HFBR-4516 with crimp ring	2104238-6	1 (bag of 20)

3- PROCEDURE

WARNING!

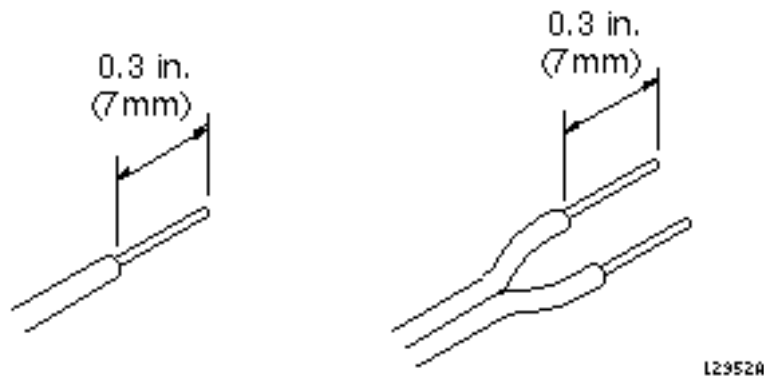
FERROUS MATERIAL HAZARD! THE CRIMP TOOL, AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE, CONTAIN FERROUS MATERIAL. THEY WILL BE STRONGLY ATTRACTED TO THE MAGNET, AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST TEN FEET AWAY FROM THE MAGNET.

1. Remove the end of the fiber optic cable to be repaired from magnet room, if possible. If this is not possible, the cable must be unstrapped from the magnet enclosure before it can be worked on; the end must be at least ten feet away from the magnet. Use extreme caution when using the crimp tool and side cutter in the magnet room; **see WARNING above.**
2. Cut cable to desired length.
3. On duplex cables, separate the two fibers approximately 2.0 inches (50 mm) back from the ends.

Note

The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable in to the connector.

Strip off approximately 0.3 inch (7 mm) of the outer jacket with the 16 gauge wire stripper. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable. See Illustration L2952A.



STRIP CABLE INSULATION
ILLUSTRATION L2952A

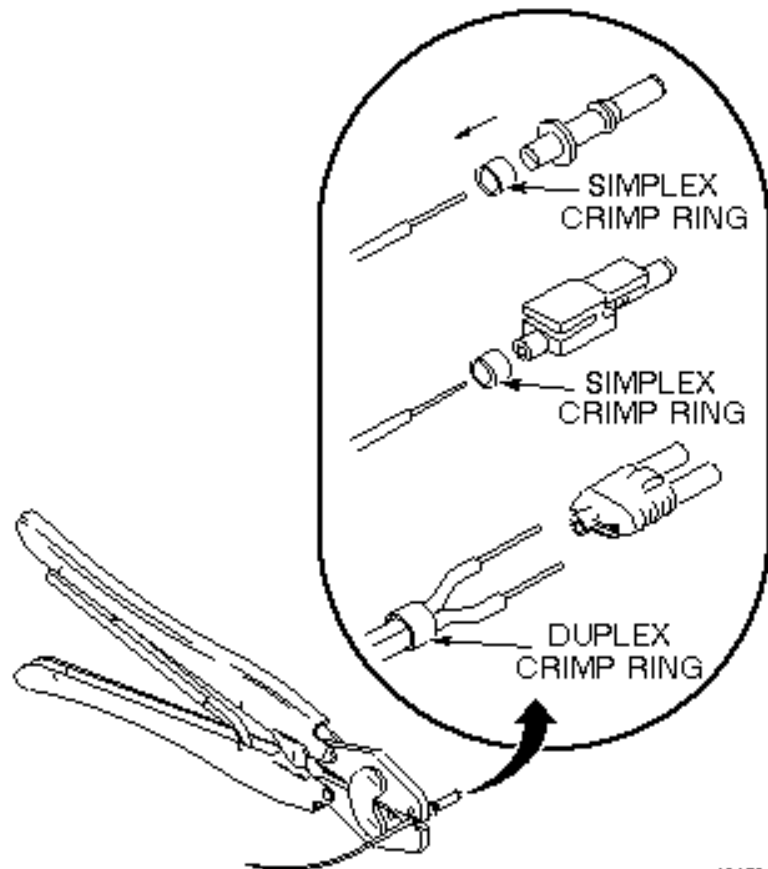
WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE, CONTAIN FERROUS MATERIAL. THEY WILL BE STRONGLY ATTRACTED TO THE MAGNET, AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST TEN FEET AWAY FROM THE MAGNET.

Note

One crimp tool is used for all connector crimping requirements

Place the crimp ring and connector over the end of the cable; the fiber should protrude about 0.12 inch (3 mm) through the end of the connector. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool (see Illustration L2953A).



NEW FIBER CONNECTOR
ILLUSTRATION L2953A

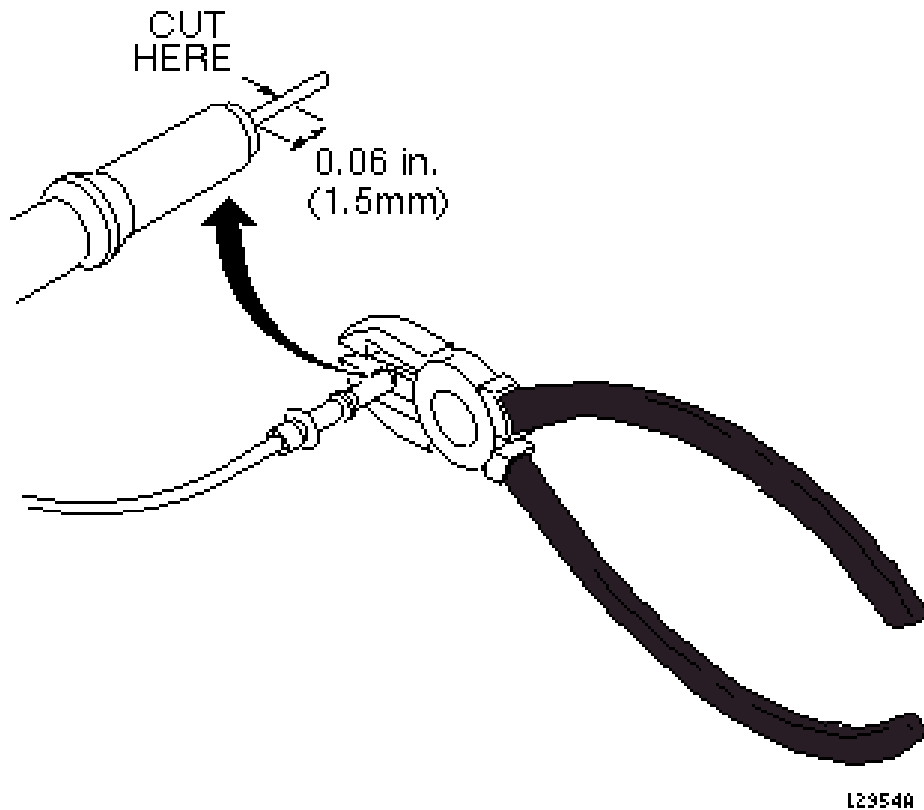
Note

The connector crimp rings cannot be exchanged between simplex and duplex connectors. The simplex crimp ring has a dull luster; the duplex crimp ring is shiny and has a thinner wall.

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

6. Cut off some of the excess fiber protruding from the connector, so the trimmed fiber extends approximately 0.06 inch (1.5mm) (see Illustration L2954A).



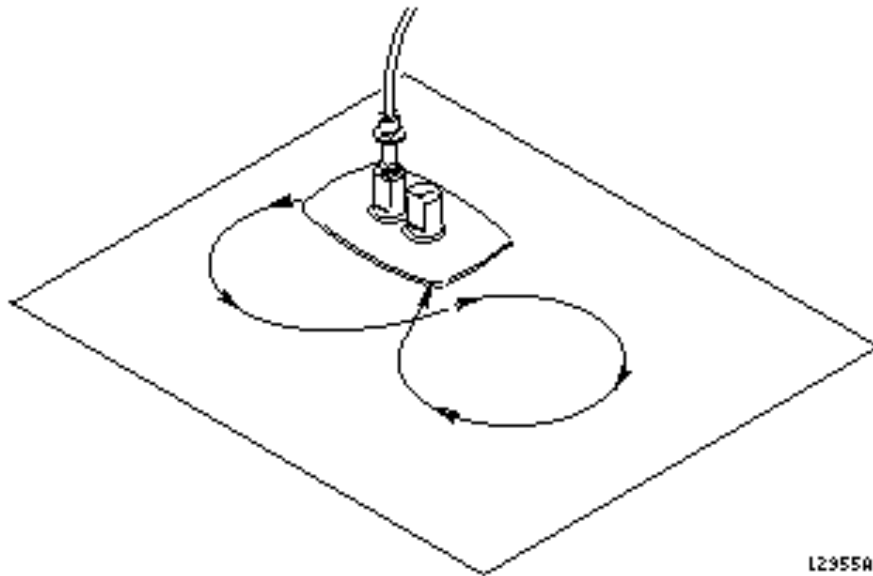
FIBER TRIMMING
ILLUSTRATION L2954A

7. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.

Note

The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible

8. Place the #600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture (see Illustration L2955A). Wipe the connector and fixture with a clean cloth or tissue.



POLISHING THE FIBER
ILLUSTRATION L2955A

9. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film, and continue to polish the fiber and connector for approximately 25 strokes. The fiber end should be flat, smooth, and clean. The cable can now be used.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.